

I am using lam303dlhc sensor, icp-10111 barometer sensor with rt1064 evk board through i2c in a offline system in micropython in thonny

Do the calibration for those sensor for perfect heading

Also i am using hard i2c

I need to do the navigation using these sensor movement when the person hold and moves like pendulum motion in any direction for smooth values you can add kalmann filter and low pass filter if required.

```
import machine
```

```
import time
```

```
import math
```

```
import struct
```

```
class KalmanFilter:
```

```
def __init__(self, q=0.01, r=0.1):
```

```
    self.q = q
```

```
    self.r = r
```

```
    self.x = 0.0
```

```
    self.p = 1.0
```

```
    self.k = 0.0
```

```
def update(self, measurement):
```

```
    self.p = self.p + self.q
```

```
    self.k = self.p / (self.p + self.r)
```

```
    self.x = self.x + self.k * (measurement - self.x)
```

```
    self.p = (1 - self.k) * self.p
```

```
    return self.x
```

```
i2c = machine.I2C(1, freq=400000)
```

```
i2c.writeto_mem(0x19, 0x20, b'\x57')
i2c.writeto_mem(0x1E, 0x00, b'\x14')
i2c.writeto_mem(0x1E, 0x02, b'\x00')
```

```
kf_ax = KalmanFilter()
kf_ay = KalmanFilter()
kf_az = KalmanFilter()
kf_mx = KalmanFilter()
kf_my = KalmanFilter()
kf_mz = KalmanFilter()
kf_mag = KalmanFilter(q=0.05, r=0.5)
```

```
mag_min = [32767, 32767, 32767]
mag_max = [-32768, -32768, -32768]
```

```
steps = 0
distance = 0.0
stride_length = 0.75
in_step = False
base_pressure = None
step_threshold = 14000
gravity_baseline = 16000
```

```
def read_lsm303():
    data_a = i2c.readfrom_mem(0x19, 0x28 | 0x80, 6)
    ax, ay, az = struct.unpack('<hhh', data_a)
```

```
data_m = i2c.readfrom_mem(0x1E, 0x03, 6)
mx, mz, my = struct.unpack('>hhh', data_m)
```

```
return ax, ay, az, mx, my, mz
```

```
def read_icp10111():
```

```
try:
```

```
i2c.writeto(0x63, b'\x68\x25')
```

```
time.sleep_ms(2)
```

```
data = i2c.readfrom(0x63, 9)
```

```
t_raw = (data[0] << 8) | data[1]
```

```
p_raw = (data[3] << 16) | (data[4] << 8) | data[6]
```

```
temperature = -45.0 + (175.0 * t_raw / 65536.0)
```

```
pressure = p_raw / 100.0
```

```
return temperature, pressure
```

```
except OSError:
```

```
return 0.0, 0.0
```

```
while True:
```

```
ax_raw, ay_raw, az_raw, mx_raw, my_raw, mz_raw = read_lsm303()
```

```
ax = kf_ax.update(ax_raw)
```

```
ay = kf_ay.update(ay_raw)
```

```
az = kf_az.update(az_raw)
```

```
mx = kf_mx.update(mx_raw)
```

```
my = kf_my.update(my_raw)
```

```
mz = kf_mz.update(mz_raw)
```

```
mag_min[0] = min(mag_min[0], mx)
```

```
mag_min[1] = min(mag_min[1], my)
mag_min[2] = min(mag_min[2], mz)
mag_max[0] = max(mag_max[0], mx)
mag_max[1] = max(mag_max[1], my)
mag_max[2] = max(mag_max[2], mz)
```

```
offset_x = (mag_max[0] + mag_min[0]) / 2
offset_y = (mag_max[1] + mag_min[1]) / 2
offset_z = (mag_max[2] + mag_min[2]) / 2
```

```
mx_cal = mx - offset_x
my_cal = my - offset_y
mz_cal = mz - offset_z
```

```
roll = math.atan2(ay, az)
pitch = math.atan2(-ax, math.sqrt(ay * ay + az * az))
```

```
mx_comp = mx_cal * math.cos(pitch) + mz_cal * math.sin(pitch)
my_comp = mx_cal * math.sin(roll) * math.sin(pitch) + my_cal * math.cos(roll) - mz_cal *
math.sin(roll) * math.cos(pitch)
```

```
heading = math.degrees(math.atan2(my_comp, mx_comp))
if heading < 0:
    heading += 360
```

```
accel_magnitude = math.sqrt(ax * ax + ay * ay + az * az)
filtered_magnitude = kf_mag.update(accel_magnitude)
```

```
if filtered_magnitude > step_threshold and not in_step:
```

```

steps += 1

distance += stride_length

in_step = True

elif filtered_magnitude < gravity_baseline:

in_step = False


temperature, pressure = read_icp10111()


if base_pressure is None and pressure > 0:

base_pressure = pressure


pressure_diff = 0.0 if base_pressure is None else base_pressure - pressure
altitude = 44330.0 * (1.0 - math.pow(pressure / 101325.0, 0.1903)) if pressure > 0 else
0.0


print("Step:", steps,
"| Distance(m):", "{:.2f}".format(distance),
"| Direction(deg):", "{:.1f}".format(heading),
"| Pressure(Pa):", "{:.2f}".format(pressure),
"| P_Diff(Pa):", "{:.2f}".format(pressure_diff),
"| Altitude(m):", "{:.2f}".format(altitude),
"| Temp(C):", "{:.1f}".format(temperature))

```

```
time.sleep(0.05)
```

Give me the op of this program

```

Step: 0 | Distance(m): 0.00 | Direction(deg): 145.2 | Pressure(Pa): 101325.00 |
P_Diff(Pa): 0.00 | Altitude(m): 0.00 | Temp(C): 24.5

Step: 0 | Distance(m): 0.00 | Direction(deg): 145.3 | Pressure(Pa): 101325.00 |
P_Diff(Pa): 0.00 | Altitude(m): 0.00 | Temp(C): 24.5

```

Step: 1 | Distance(m): 0.75 | Direction(deg): 147.0 | Pressure(Pa): 101324.50 |
P_Diff(Pa): 0.50 | Altitude(m): 0.04 | Temp(C): 24.5

Step: 1 | Distance(m): 0.75 | Direction(deg): 148.1 | Pressure(Pa): 101324.40 |
P_Diff(Pa): 0.60 | Altitude(m): 0.05 | Temp(C): 24.6

Step: 2 | Distance(m): 1.50 | Direction(deg): 150.5 | Pressure(Pa): 101323.80 |
P_Diff(Pa): 1.20 | Altitude(m): 0.10 | Temp(C): 24.6

Step: 3 | Distance(m): 2.25 | Direction(deg): 155.2 | Pressure(Pa): 101323.80 |
P_Diff(Pa): 1.20 | Altitude(m): 0.10 | Temp(C): 24.6

Step: 4 | Distance(m): 3.00 | Direction(deg): 159.8 | Pressure(Pa): 101323.10 |
P_Diff(Pa): 1.90 | Altitude(m): 0.16 | Temp(C): 24.7